

Exam Number: _____

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B.E. 8th (REG/ ATKT) EXAMINATION APRIL 2025

Subject Name: Next Generation Networks

Subject Code: CE801-N

Date: 05/04/2025(Saturday)

Time: 12.30 pm to 03.30 pm

Total marks: 70

Instructions:

1. Answer each section in separate Answer sheet.
2. All questions are compulsory.
3. Indicate clearly, the options you attempt along with its respective question number.
4. Use the last page of main supplementary for rough work.

Section-I

- Q.1(A) What is Multiplexing technique? What is the need of it? (5)
- Q.1(B) Explain Generations of Mobile Communication Technology. (5)
- Q.1(C) Explain CDMA, TDMA, FDMA and SDMA. (5)

OR

- Q.1(C) Illustrate GPRS System Architecture. (5)

- Q.2 (A) Draw and Explain Architecture and Services of GSM. (5)
- Q.2(B) Explain Reactive Routing Protocols – DSR, AODV. (5)

OR

- Q.2 (A) Describe GSM Addresses and Identifiers. (5)
- Q.2(B) Draw and Explain Architecture of ~~UTMS~~ ^{UMTS}. (5)

- Q.3 (A) Draw the Architecture of Mobile IP & Explain working of it in detail. (5)
- Q.3 (B) What is Proactive Protocol? Explain DSDV in detail. (5)

OR

- Q.3 (A) Differentiate MANET and VANET. (5)
- Q.3 (B) Explain Mobile TCP in detail. (5)

Section-II

- Q.4 (A) Explain the Architecture and working of WAP. (5)
- Q.4 (B) Explain WDP & WTLS in detail. (5)
- Q.4 (C) How does SDN differ from Traditional Network? Draw and Explain the Architecture of SDN. (5)

OR

- Q.4 (C) What are the special constraints and requirements of mobile device operating systems? (5)
- Q.5 (A) What are the major evolving network requirements leading to SDN adoption? (5)
- Q.5 (B) Explain the key functions of the SDN Data Plane and its role in network architecture. (5)

OR

- Q.5 (A) Compare SDN and NFV. Briefly Explain NFV. (5)
- Q.5 (B) Explain the OpenFlow Protocol and how it enables communication between SDN controllers and switches. (5)

- Q.6 (A) What is the Southbound Interface in SDN? How does it facilitate communication? (5)
- Q.6 (B) What is the Network Services Abstraction Layer in SDN, and why is it important? (5)

OR

- Q.6 (A) What is the ITU-T Model in SDN, and how does it standardize network control? (5)
- Q.6 (B) Discuss the application of SDN in mobility and wireless networks. (5)
